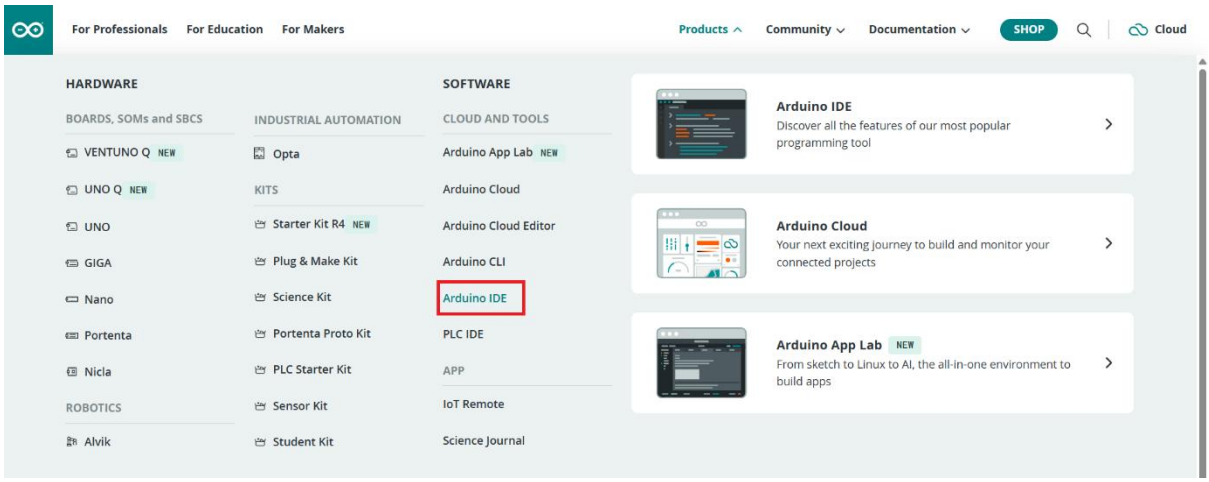
Panduan Arduino IDE untuk Waveshare ESP32 S3 Zero

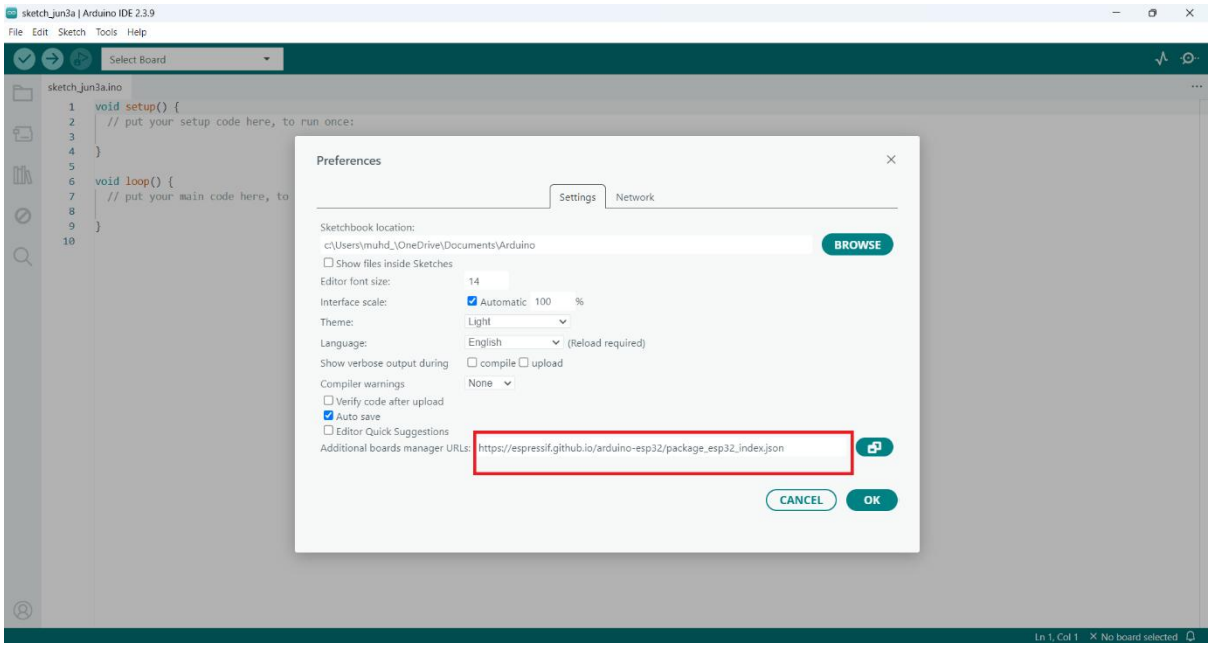
1. Search di google Arduino dan klik https://www.arduino.cc/



2. Pada main page klik product dan pilih Arduino IDE di bahagian software.

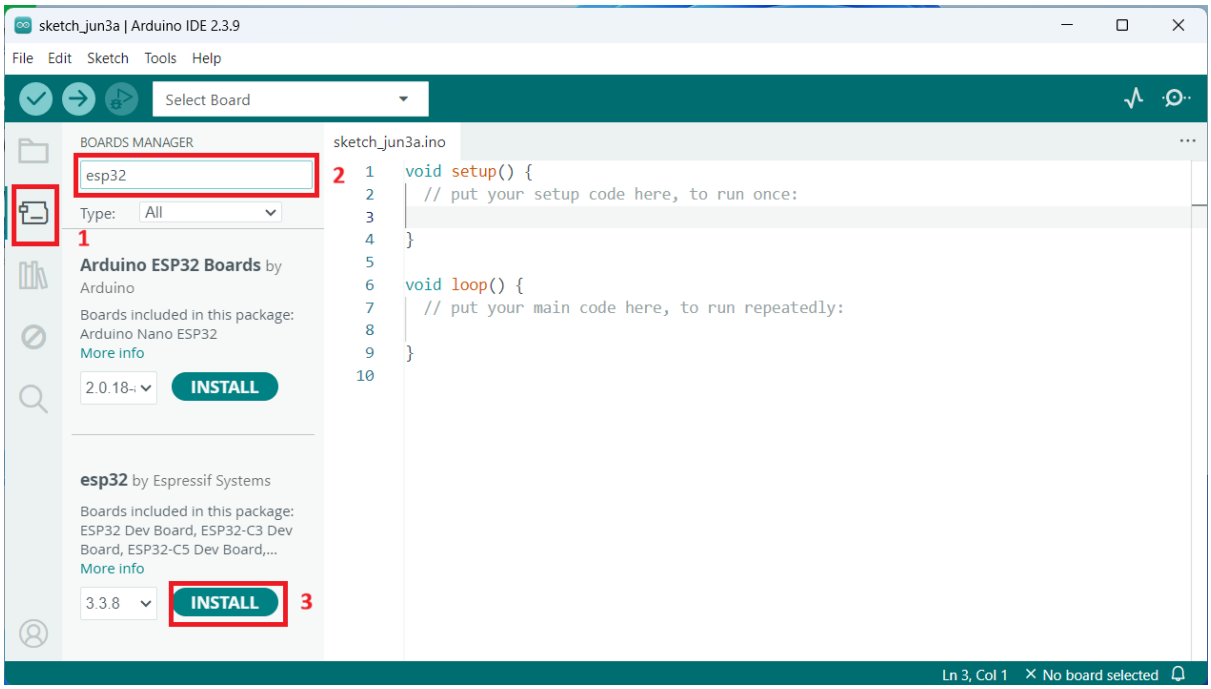


3. Download Arduino IDE 2.X.X seperti gambar rajah di atas.
4. Install Arduino IDE di laptop.
5. Sebelum Waveshare ESP32 S3 Zero boleh digunakan dalam Arduino IDE, anda perlu install ESP32 board configuration.

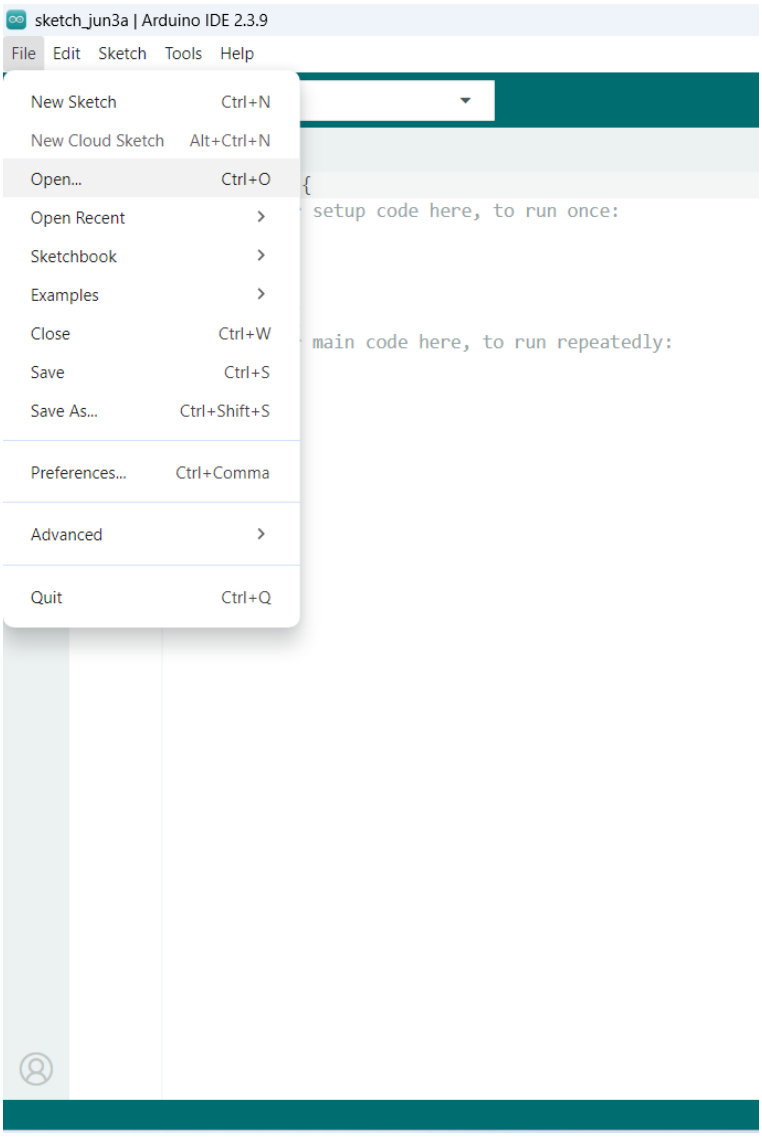


6. Klik "File" -> "Preferences", cari "Additional Board Manager URLs" dalam Settings screen, copy & paste link di bawah dan klik ok.

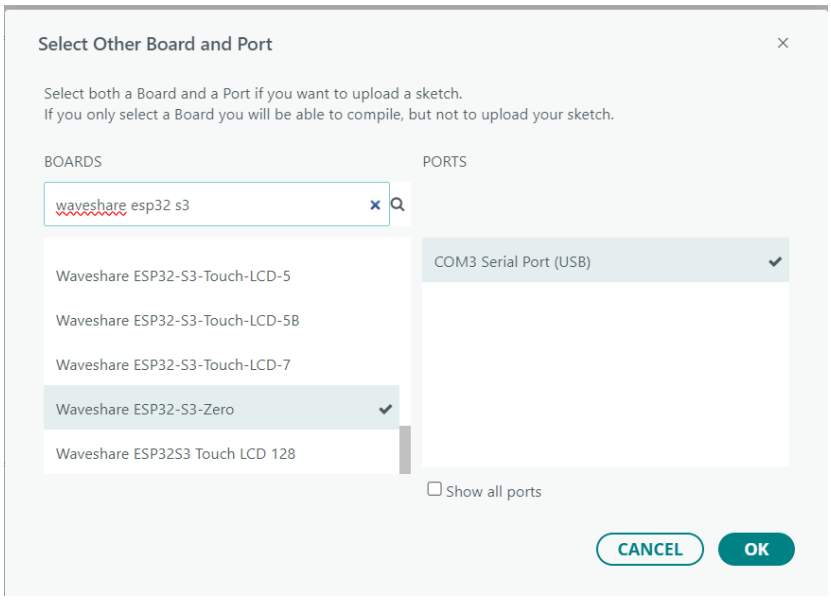
   https://espressif.github.io/arduino-esp32/package_esp32_index.json



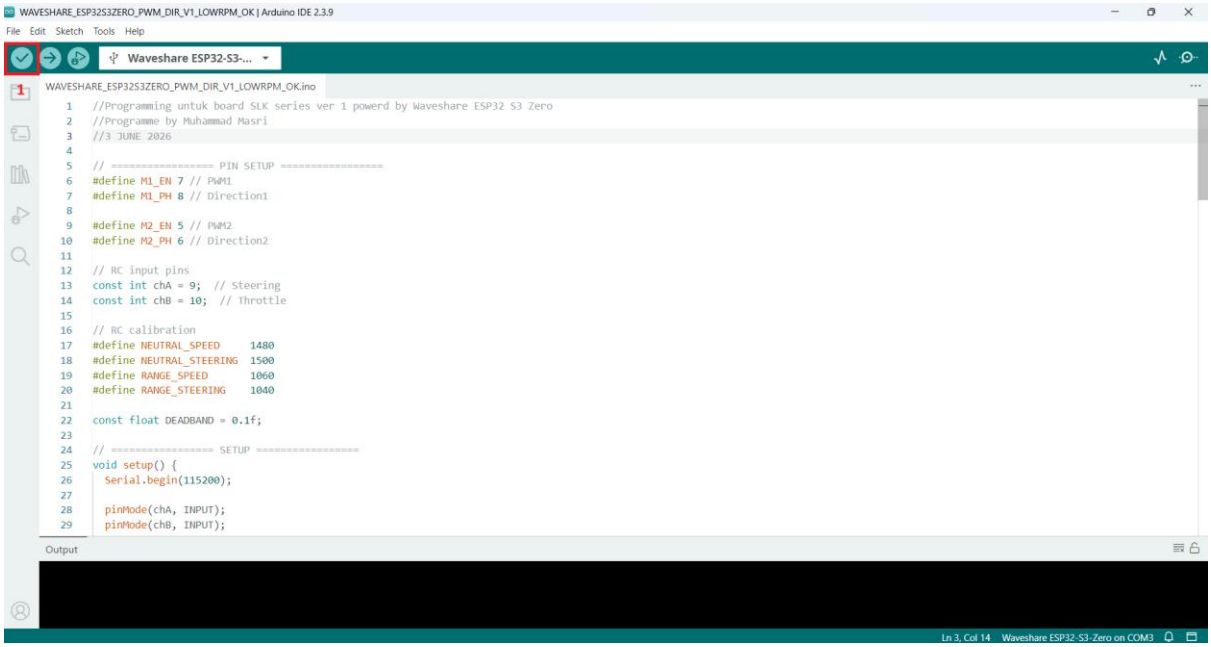
7. Kemudian ikuti langkah di atas iaitu **1. Klik "Board Manager"**. **2. Taip "ESP32"**. **3. Pilih esp32 by Espressif System dan install**. Selepas install, restart semula Arduino IDE.



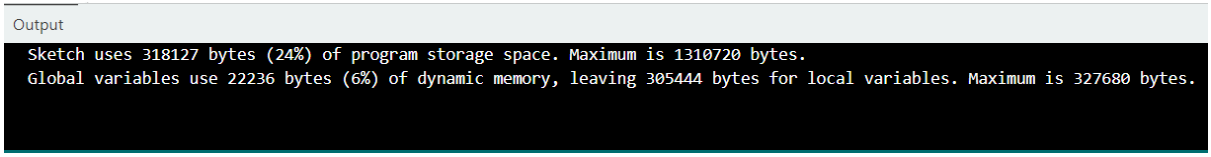
8. Klik file > open dan pilih file program yang telah diberi secara softcopy



9. Sambungkan Waveshare ESP32 ke laptop dengan menggunakan cable type C. Kemudian klik Select board > Select Other Board and Port. Taip waveshare esp32 s3 dan cari board Waveshare ESP32-S3-Zero.



10. Untuk pertama kali cuba klik verify button untuk compile programme dan memeriksa sebarang error.



11. Jika compile programme berjaya akan mengeluarkan output seperti diatas.

```
void applyCalibration(float speedPercent, float steeringPercent, int &leftSpeed, int &rightSpeed) {
  if (speedPercent > 0.0f && steeringPercent == 0.0f) {
    Serial.println("FWD");
    //CALIBRATION SEMASA ROBOT FORWARD
    //jika robot bergerak senget ke arah kanan semasa throttle sahaja ditekan,
    //kurangkan nilai pada leftspeed. Maximum value 1.0,
    //kurangkan nilai contohnya 0.95 dan upload semula coding.
    leftSpeed *= 1.0f;  // Adjust for motor imbalance
    rightSpeed *= 1.00f;
  } else if (speedPercent < 0.0f && steeringPercent == 0.0f) {
    Serial.println("BWD");
    //CALIBRATION SEMASA ROBOT BACKWARD
    leftSpeed *= 1.0f;
    rightSpeed *= 1.0f;
  } else if (steeringPercent > 0.0f) {
    Serial.println("RIGHT");
    //CALIBRATION SEMASA ROBOT TURN RIGHT
    leftSpeed *= 1.0f;
    rightSpeed *= 1.0f;
  } else if (steeringPercent < 0.0f) {
    Serial.println("LEFT");
    //CALIBRATION SEMASA ROBOT TURN LEFT
    leftSpeed *= 1.0f;
    rightSpeed *= 1.0f;
  }
}
```

12. Akhir sekali, jika robot bergerak senget semasa throttle sahaja ditekan, nilai parameter line pd 107 & 108 boleh diubah. Kemudian sila klik upload untuk memuat naik programme terkini pada Waveshare ESP32 S3 Zero.

13. Parameter semasa backward juga boleh diubah jika robot senget semasa reverse.